A 65-nm CMOS Millimeter-Wave Ear-Worn Non-Invasive Continuous Glucose Monitoring Chipset Using a 0.17mm^2 72mW Transmitter and a 0.75mm^2 80mW Direct-Conversion Receiver

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***Abstract*— Diabetes necessitates monitoring of blood glucose levels. However, the existing methods are invasive in nature while being expensive and not being suitable for continuous monitoring. Thus, this study proposed a low-cost millimeter-wave (MMW) complementary metal-oxide-semiconductor (CMOS) transmitter and receiver chipset to facilitate ear-worn non-invasive continuous glucose monitoring for diabetic care. The direct conversion receiver method was adopted, which reduced the total area of the entire circuit by 37% compared to the conventional technology, thus decreasing the cost. A prototype chip of the transmitter and receiver circuits was designed and developed in 65-nm CMOS, and the MMW transmission/reception operation was confirmed in the 30-GHz band between two simulated human bodies composed of DI water and glucose.**

Keywords—blood glucose monitoring, non-invasive, CMOS, millimeter wave, low-cost, low-power

I. Introduction

In recent years, the increase in the diabetic population has become a major social problem [1]. Diabetes is difficult to diagnose, and late diagnosis increases the risk of complications such as retinopathy, nephropathy, and stroke. To reduce this risk, the variability in blood glucose levels must be reduced; hence, self-monitoring of blood glucose levels daily using an inexpensive blood glucose meter that enables continuous evaluation is crucial.

Maintaining blood glucose levels as close to the normal range as possible and keeping the body healthy despite diabetes prevents the onset of symptoms and complications [2],[3]. However, blood glucose monitoring, which is currently the primary method used, is invasive, physically and economically burdensome, and does not facilitate continuous evaluation [4],[5].

As an alternative, this study proposed blood glucose monitoring using millimeter-wave (MMW) technology, as shown in Fig. 1. In this method, blood glucose levels were measured by placing a transmitter and receiver between the thin skin of the patient's earlobe or finger. Considering the comfort of patients, a clip-shaped device was assumed for adoption in practical applications. This method is less burdensome for patients and enables continuous evaluation, rendering it suitable for the prevention and treatment of diabetes. Against this background, this study aimed to develop and evaluate a complementary metal-oxide-semiconductor CMOS miniature MMW transmitter/receiver circuit that enables noninvasive and continuous wearable blood glucose sensing. To the best of our knowledge, there are no reports on such measurements. Thus, our results are the world's first demonstration of a CMOS MMW transmitter/receiver circuit for continuous blood glucose measurements.

Among the many noninvasive methods being considered, electromagnetic (EM) waves are promising because they can directly penetrate biological tissues [6],[7]. Various glucose-sensing systems have been developed using microwaves [8]–[12], infrared (IR) waves [13],[14], and light waves [15],[16]. However, as blood glucose levels vary only in the range of milligrams per deciliter (mg/dL), the measurement of glucose concentration with a noninvasive, small, and wearable device using any sensor technology, including ultrasound and optics, is challenging. All these concepts are based on a concentration-dependent dielectric constant.

Moreover, most techniques and systems have low sensitivity, and are not applicable to lossy aqueous solutions. Further, they do not demonstrate a clear correlation between

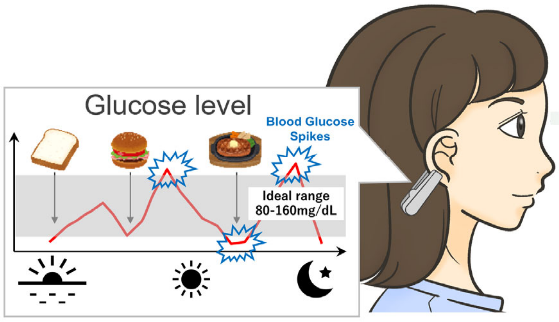


Fig. 1. Conceptual image of glucose monitoring with the proposed system.

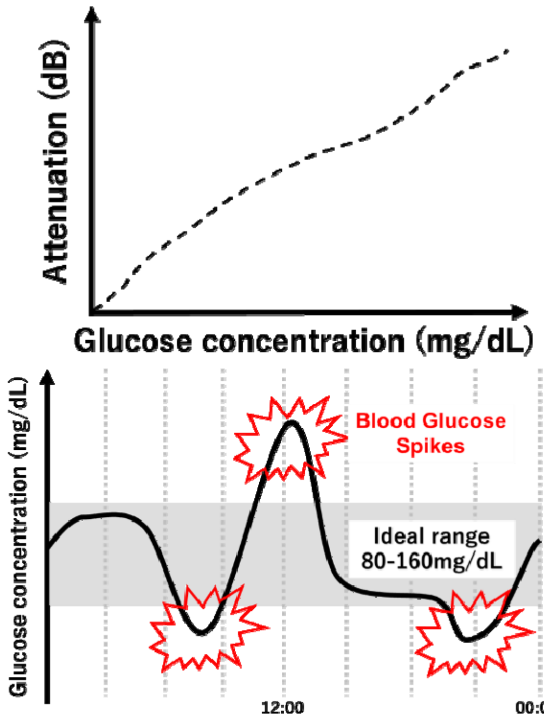


Fig. 2. Evaluation of blood glucose levels.

measurement results and changes in glucose concentration. Because EM fields can penetrate different tissues, such as skin and fat, various MMW measurement techniques have been introduced in the past [17]–[29]. EM waves exploit the fact that the glucose concentration in human blood affects its EM properties (dielectric constant and loss tangent). Consequently, a promising correlation between the measured electromagnetic properties of blood and its glucose concentration has been identified [22]. The characterization of MMWs can be performed in free space and does not require precise tuning, as is the case for optical frequencies. Furthermore, the MMW band is ideal for characterizing composite materials, where the homogeneity scale is of the order of millimeters [30]. However, the challenge with detection using MMW reflections is that the signal rarely reaches the blood, and noise may not provide sufficient information to detect small changes in glucose levels.

However, most techniques and systems have low sensitivity, are not applicable to lossy aqueous solutions, and do not demonstrate a clear correlation between measurement results and changes in glucose concentration. Because electromagnetic fields can penetrate different tissues, such as skin and fat, various MMW measurement techniques have been introduced in the past [17]–[29]. EM waves are based on the fact that the glucose concentration in human blood affects its electromagnetic properties (dielectric constant and loss tangent), and a promising correlation between the measured electromagnetic properties of blood and its glucose concentration has been identified [22]. A part of this study was based on a student design contest proposal at a conference [46].

II. Detection Method

According to the bi-relaxation process, the relative dielectric constant of the dispersive molecules is complex and frequency dependent [33].

Various techniques for measuring the change in dielectric properties of the material under test (MUT) have been proposed in the past to measure the change in dielectric properties of glucose [20]–[23]. In the case of a two-component aqueous solution, three parameters are used: the electrostatic permittivity at low frequency, dielectric constant at high frequency, and characteristic relaxation time of the medium parameters. The dielectric constant ε can be approximated as a function of the angular frequency and the concentration of the hydrophilic substance, as shown in Equation (1) below:

$$\gamma = \alpha + j\beta, \tag{1}$$

where ω is the angular frequency and χ is the glucose concentration [20]. α and β are expressed as:

$$\alpha = \frac{\omega\sqrt{\varepsilon'_r(\omega,X)}}{c_0}\sqrt{\frac{1}{2}\sqrt{1+\left(\frac{\varepsilon''_r(\omega,X)}{\varepsilon'_r(\omega,X)}\right)^2}-1} \tag{2}$$

$$\beta = \frac{\omega\sqrt{\varepsilon'_r(\omega,X)}}{c_0}\sqrt{\frac{1}{2}\left(1+\sqrt{1+\left(\frac{\varepsilon''_r(\omega,X)}{\varepsilon'_r(\omega,X)}\right)^2}\right)}. \tag{3}$$

A previous study confirmed through measurements in a two-component aqueous solution at the Kα-band that the calculated and measured results were reliable. Further, it was reported that the dielectric constant ε_r changed linearly with the angular frequency ω of the electric field and the glucose concentration χ in the blood [21]. Based on this data, sensing can be achieved by reading the changes in the received signal strength in response to changes in blood glucose concentration. When the MMW transmitted from the transmitter passes through the skin, its dielectric absorption coefficient increases with increase in the blood glucose level,

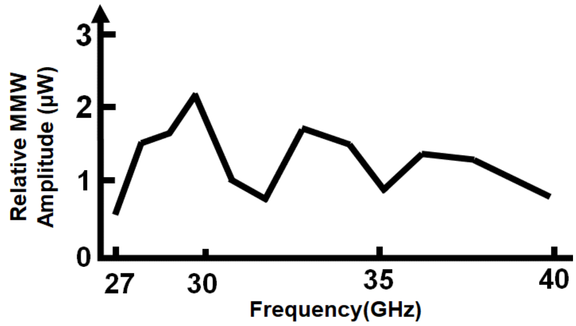


Fig. 3. Frequency response of blood glucose concentration.

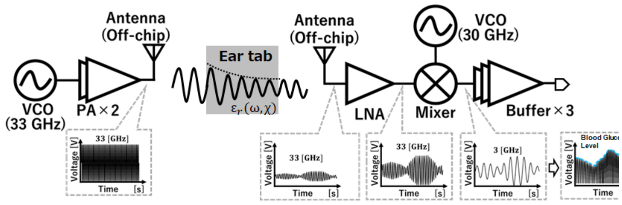
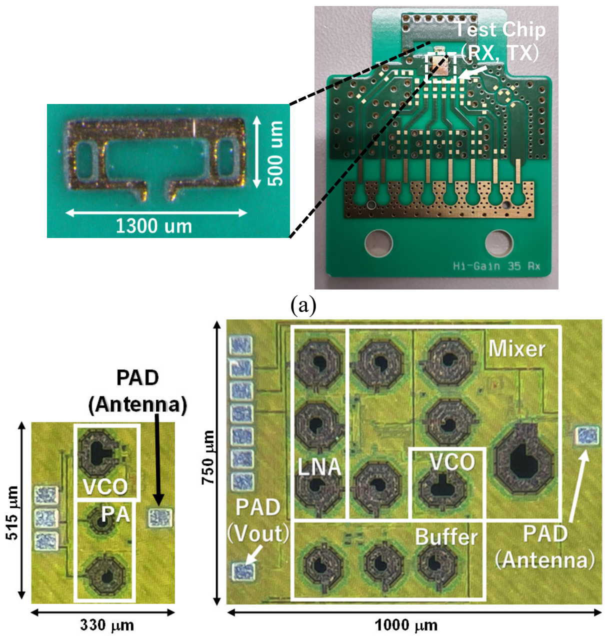


Fig. 4. Operation principle of blood glucose sensor.



(b) (c)

Fig. 5. (a) Evaluation board and antenna. (b) Transmitter die photograph. (c) Receiver die photograph.

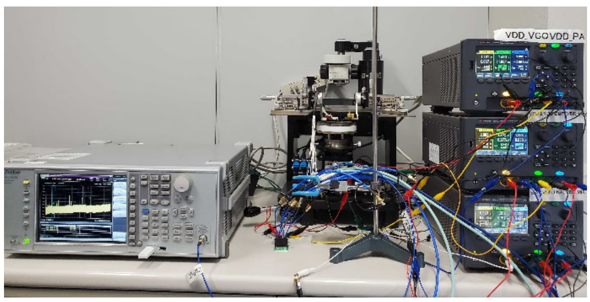


Fig. 6. Experimental setup.

whereas the received signal strength decreases. The S_21 dB at the two ports is shown in Equation (4), with the real part α contributes to the attenuation in Equation (1):

$$S_{21} = 20\, log \frac{V_{out}}{V_{in}} = -\alpha \cdot d, \qquad (4)$$

where d represents the distance. Moreover, the larger the distance, the larger the attenuation and the thinner the skin between the earlobes and fingers. As shown in Fig. 2, the relative value of blood glucose can be determined by sensing the received signal strength. This study aimed to detect blood glucose spikes by continuously evaluating the variation in relative blood glucose values, which has the advantage of being resistant to the effects of external electrical and thermal noise.

The challenge with detection using reflections is that the signal barely reaches the blood. Consequently, this minimal interaction may provide insufficient information to detect small changes in glucose concentration. Therefore, this study aimed to observe transmitted waves. To utilize the transmitted data, a sensor operating in the Kα-band was developed to non-invasively detect glucose levels in the blood. There exists a positive correlation between the blood glucose level and the MMW transmission from the rat ear, and the signal change greatly exceeded the noise of the system [31],[32]. The device was based on a waveguide and was tested on human tissues (2 mm). The Kα-band frequency range does not have any correlation with components other than glucose in plasma. Moreover, it correlates with the metabolic rate of the human body when only blood glucose concentration is manipulated in vitro with a 10–15 min delay [31].

Figure 3 shows the measurement results of the earlobes of mice from previous studies [31],[32]. The targeted frequency range was within the range of experiments conducted on human subjects in previous studies [32]. Within this range, we selected 30 GHz, which exhibited the highest relative MMW amplitude, as a tradeoff between the correlation with each frequency and the attenuation rate. This resulted in high sensing accuracy. The expected signal attenuation S_{21} was 20–30 dB [34].

III. Sensor Implementation, Chip Prototyping, and Measurements

The sensor was designed at 30 GHz using the TSMC 65-nm CMOS standard process. The overall architecture is shown in Fig. 4. The prototype chip is shown in Fig. 5, including the PADs, with areas of 0.75 and 0,17 mm^2 for the receiver and transmitter, respectively. The total area of the designed circuits was reduced by approximately 37% compared with that of the circuits in previous studies [34]. The measurement system, wherein the input and output parts of the transmitter and receiver circuits were connected to the antenna by bonding wires, is shown in Fig. 5(a). Although a large test board was used for the actual measurement, the total area of the antenna and transmitter/receiver circuit was only 3 mm^2, which is sufficiently small for mounting as a wearable device.

In the fabricated transmitter/receiver circuit shown in Fig. 4, the supply voltage to each PAD was 2–2.03 V for the VCO and 1.2–1.23 V for the others. The experimental setup for these measurements is shown in Fig. 6. The spectrum analyzer used was an Anritsu MS2830A, and the power supply was a KEYSIGHT E36312A.

As shown in Fig. 7, the distance between the transmitter and receiver was 2 mm, which is sufficient to pinch the skin of the human body. This approximation is valid because water accounts for a high percentage (approximately 50%)

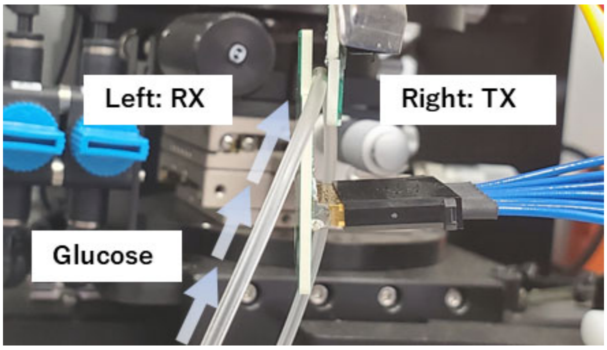


Fig. 7. Enclosed view of measurement setup.

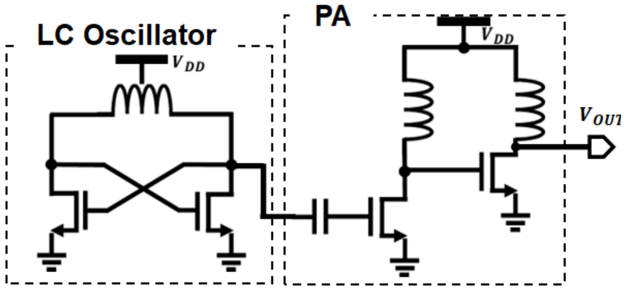


Fig. 8. Transmitter circuit diagram.

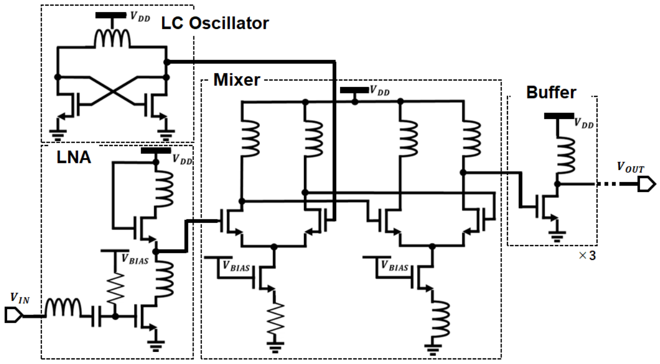


Fig. 9. Receiver circuit diagram.

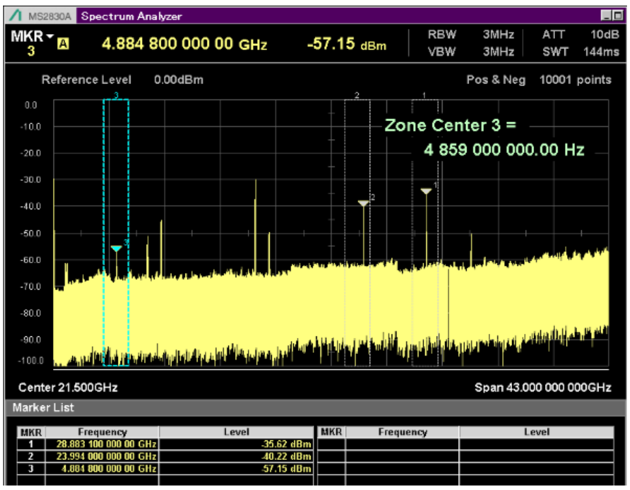


Fig. 10. Receiver output spectrum.

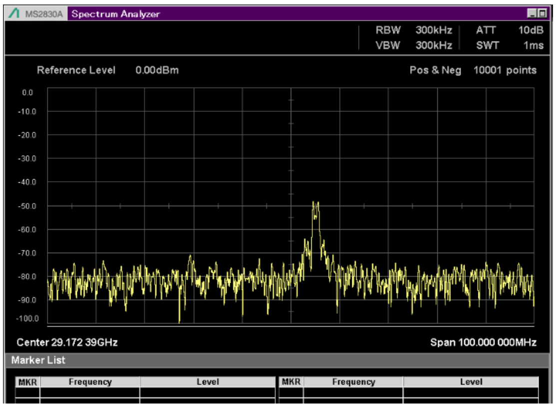


Fig. 11. Enlarged view of waveform after down-conversion.

of the total volume of human blood and contains other important constituents (glucose, Na, Ca, K, and Cl) in different proportions. Because blood is a liquid composed of a considerable amount of water, a glucose solution was considered instead of a skin layer in this study. Further, polyvinyl chloride, a DEHP-proof PVC material from Covidien Japan, was used. Glucose solutions of different concentrations (DI water + D-glucose) were passed through the infusion tube, and the receiver output was measured for a constant intensity of the transmission signal strength. The outer diameter of the infusion tube was 2 mm, which was sufficient for the size of the antenna. The glucose concentration varied between 90, 180, and 540 mg/dL.

A. Transmitter circuit

The designed circuit is illustrated in Fig. 8. The circuit configuration comprised an LC oscillator oscillating at a frequency of 30 GHz and a power amplifier amplifying the output power by more than 20 dB. From the measurement results shown in Fig. 10, the phase noise of the transmitter observed by the receiver was −101.1 dBc/Hz at an offset frequency of 1 MHz. The power consumption of the transmitter was 72 mW.

B. Receiver circuit

A direct-conversion system was used as the receiver circuit and was expected to achieve a small form factor. Figure 9 shows a schematic of the designed circuit. The first stage of the LNA amplified the received signal to drive the mixer efficiently.

The phase of the local LC oscillator output was adjusted by changing the supply voltage of the LC oscillator to align the phase of the input signal from the antenna and the local LC oscillator on the receiver side.

From the results presented in Fig. 10, the down-converted frequency component at 29.30 – 23.98 GHz = 5.32 GHz was confirmed. Regarding the circuit performance, for a high signal-to-noise ratio (SNR), the value of ΔS_{21} that can be measured accurately is small. Concurrently, operating at a higher frequency results in a lower SNR value because of the higher signal attenuation in electrically long samples. Consequently, the uncertainty of the detector, that is, the minimum value of the sensing accuracy that can be measured reliably (i.e., ΔS_{21}), also increases with frequency. The output of the receiver circuit is shown in Fig. 11. The results show that the SNR was greater than 30, which confirmed the sensing capability of the receiver. The power consumption of the receiver was 80 mW.

IV. Comparison

A comparison between the receiver circuit and circuit configuration fabricated in a previous study is presented in Table 1. The total area was reduced from 1.459 to 0.92 mm^2, corresponding to a 37% reduction. In previous studies, the

TABLE I Comparison with Previous Studies

Publication	Sensors 2016 [12]	IEEE Access 2020 [17]	IEEE TBME 2018 [25]	IEEE IRMMW 2015 [34]	This Work
Process				65nm CMOS	65nm CMOS
Concentration (mg/dL)	30–8000	70–120	0–300	-	90–540
Sample volume (dL)	-	0.1	-	-	0.1
Frequency (GHz)	7.5	50–70	60–80	33–37	29.3–30.0
Sensitivity (dB per mg/ml)	0.008	0.8-1.0	0.23	-	0.17
Size	Large	Large	Large	Antenna: 0.96 mm^2 TX: 0.595 mm^2 RX: 0.864 mm^2	Antenna: 0.65 mm^2 TX: 0.17 mm^2 RX: 0.75 mm^2
Power	N/A	N/A	N/A	N/A	TX: 72 mW RX: 80 mW
Used model	In-vitro (Micro-fluidic sensor)	In-vitro (Blood test tube)	In-vitro (Plastic liquid container)	No measure-ment results	In-vitro (Blood test tube)
Linearity	0.99	-	-	-	0.95

size was set to be large for sensing methods using a waveguide as the measurement system. However, previous studies did not show the evaluation data of the actual measured values using the prototype chip; thus, our study is the first to confirm that the signal attenuation S_{21} changes according to the glucose concentration using the prototype chip as an MMW-applied transmitter/receiver circuit for blood glucose monitoring. This is the first study to demonstrate that a CMOS chip detects signal attenuation S_{21} changes with glucose concentration.

V. Conclusion

This study proposed a low-cost CMOS MMW receiver circuit for non-invasive continuous glucose monitoring. A test chip was fabricated in the TSMC 65-nm standard CMOS process to satisfy the performance requirements of a previous study, and the simulation results were analyzed. The area of the fabricated chip was successfully reduced by 37% compared to that of the area of the receiver circuit fabricated in a previous study, resulting in a lower cost. We read the change in the output according to the glucose concentration in the prototype connected to the transmitter and receiver circuits.

In future, we will conduct a first-in-human clinical trial. By combining other non-invasive glucose monitoring techniques [35-45], further improvement in accuracy is expected.

Acknowledgements

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